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Inventor(s) Hodge; Chad Michael et al.

DEVICES AND METHODS FOR MAINTAINING RELIEF VALVES IN OPEN POSITIONS

Abstract

A relief valve has a closed position with a lid of the relief valve closing an opening of a storage tank and an open position with the lid pivoted away from the opening of the storage tank. A hand saver device for maintaining the relief valve in the open position includes a body comprising a U-shaped base plate having a central axis, a first end, a second end opposite the first end, a pair of lateral sides extending axially from the first end to the second end, and a recess extending axially from the first end. In addition, the hand saver device includes a pair of laterally spaced extension arms extending upwardly from the base plate. Each extension arm includes a throughbore extending laterally therethrough. Further, the hand saver device includes an elongate locking rod configured to be removably and slidably disposed in the throughbores of the extension arms.

Inventors: **Hodge; Chad Michael (Lovington, NM), Rodriguez; Casey (Lovington, NM), Pitts; Darell Eric (Lovington, NM), Zarate; Sylvester (Lovington, NM)**

Applicant: **EOG Resources, Inc.** (Houston, TX)

Family ID: **1000008489274**

Assignee: **EOG Resources, Inc. (Houston, TX)**

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims benefit of U.S. provisional patent application Ser. No. 63/562,355 filed Mar. 7, 2024, and entitled “Devices and Methods for Maintaining Relief Valves in Open Positions,” which is hereby incorporated herein by reference in its entirety for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT [0002] Not applicable.
BACKGROUND
[0003] The disclosure relates generally to relief valves. More particularly, the disclosure relates to devices and methods for maintaining emergency pressure relief valves open during inspections, maintenance, repair, or combinations thereof.
[0004] Pressure vessel systems typically include a pressure vessel defining an internal cavity in which materials such as fluids (e.g., hydrocarbon fluids) may be stored for a desired period of time. Pressure vessels typically have a designed maximum internal pressure that may not be exceeded without potentially jeopardizing the operability (e.g., the structural integrity) of the pressure vessel. To ensure the fluid pressure within internal cavity of the pressure vessel remains below this maximum operating pressure, pressure vessel systems often include a specialized pressure relief valve sometimes referred to as an emergency pressure relief valve (EPRV). Typically, an EPRV of a storage system includes a cover or lid that may be pivoted between a closed position blocking and sealing an opening in the pressure vessel to prevent fluid communication between the internal cavity of the pressure vessel and the surrounding environment, and an open position spaced apart from the opening in the pressure vessel to permit fluid communication between the internal cavity of the pressure vessel and the external environment. The lid is biased to the closed position, but is configured to transition to the open position in response to fluid pressure within the internal cavity of the pressure vessel exceeding the maximum operational pressure of the pressure vessel, thereby preventing the fluid pressure within the internal cavity of the pressure vessel from exceeding the maximum operational pressure of the pressure vessel.
BRIEF SUMMARY OF THE DISCLOSURE
[0005] Embodiments of devices for maintaining a relief valve in an open position are disclosed herein. The relief valve has a closed position with a lid of the relief valve closing an opening of a storage tank and the open position with the lid pivoted away from the opening of the storage tank. In one embodiment, a hand saver device for maintaining a relief valve in an open position comprises a body comprising a U-shaped base plate having a central axis, a first end, a second end opposite the first end, a pair of lateral sides extending axially from the first end to the second end, and a recess extending axially from the first end. In addition, the hand saver device comprises a pair of laterally spaced extension arms extending upwardly from the base plate. Each extension arm includes a throughbore extending laterally therethrough. Further, the hand saver device comprises an elongate locking rod configured to be removably and slidably disposed in the throughbores of the extension arms.
[0006] Embodiments of methods for maintaining a relief valve in an open position are also disclosed herein. The relief valve includes a pivot arm having a first end and a second end, a weight attached to the pivot arm proximal the first end, and a lid coupled to the pivot arm proximal the first end. In one embodiment, a method for maintaining a relief valve coupled to a storage tank in

an open position comprises (a) coupling a hand saver device to the relief valve with the relief valve in a closed position. The hand saver device comprises a body comprising a base plate and a pair of extension arms extending upwardly from the base plate. The base plate has a central axis, a first end, a second end opposite the first end, a pair of lateral sides extending axially from the first end to the second end, and a recess extending axially from the first end. The extension arms are disposed on opposite lateral sides of the recess. Each extension arm includes a throughbore. The hand saver device also comprises an elongate locking rod configured to be removably disposed in the throughbores of the extension arms. In addition, the method comprises (b) raising the lid to transition the relief valve from the closed position to the open position and move the second end of the pivot arm downward through the recess of the base plate after (a). Further, the method comprises (c) advancing a locking rod through the throughbores of the extension arms after (b) to prevent the second end of the pivot arm from rotating upward through the recess.
[0007] Embodiments described herein comprise a combination of features and characteristics intended to address various shortcomings associated with certain prior devices, systems, and methods. The foregoing has outlined rather broadly the features and technical characteristics of the disclosed embodiments in order that the detailed description that follows may be better understood. The various characteristics and features described above, as well as others, will be readily apparent to those skilled in the art upon reading the following detailed description, and by referring to the accompanying drawings. It should be appreciated that the conception and the specific embodiments disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes as the disclosed embodiments. It should also be realized that such equivalent constructions do not depart from the spirit and scope of the principles disclosed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] For a detailed description of various exemplary embodiments, reference will now be made to the accompanying drawings in which:
[0009] FIG. **1** is a partial perspective view of an embodiment of a fluid storage vessel in accordance with principles disclosed herein;
[0010] FIG. **2** is an enlarged, partial perspective side view of the fluid storage vessel of FIG. **1** illustrating the pressure relief valve;
[0011] FIG. **3** is a perspective view of the hand saver device of FIG. **1**;
[0012] FIG. **4** is a top view of the hand saver device of FIG. **3**;
[0013] FIG. **5** is a top view of the body of the hand saver device of FIG. **3**;
[0014] FIG. **6** is a side view of the body of the hand saver device of FIG. **3**;
[0015] FIG. **7** is a front view of the body of the hand saver device of FIG. **3**;
[0016] FIG. **8** is a front view of the locking rod of the hand saver device of FIG. **3**; and
[0017] FIG. **9** is a perspective view of the hand saver device of FIG. **3** coupled to the pressure relief valve of the fluid storage vessel of FIG. **1**.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
[0018] The following discussion is directed to various exemplary embodiments. However, one skilled in the art will understand that the examples disclosed herein have broad application, and that the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to suggest that the scope of the disclosure, including the claims, is limited to that embodiment.
[0019] Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish

between components or features that differ in name but not function. The drawing figures are not necessarily to scale. Certain features and components herein may be shown exaggerated in scale or in somewhat schematic form and some details of conventional elements may not be shown in interest of clarity and conciseness.

[0020] Unless the context dictates the contrary, all ranges set forth herein should be interpreted as being inclusive of their endpoints, and open-ended ranges should be interpreted to include only commercially practical values. Similarly, all lists of values should be considered as inclusive of intermediate values unless the context indicates the contrary.

[0021] In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . . ” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct engagement between the two devices, or through an indirect connection that is established via other devices, components, nodes, and connections. In addition, as used herein, the terms “axial” and “axially” generally mean along or parallel to a particular axis (e.g., central axis of a body or a port), while the terms “radial” and “radially” generally mean perpendicular to a particular axis. For instance, an axial distance refers to a distance measured along or parallel to the axis, and a radial distance means a distance measured perpendicular to the axis. As used herein, the terms “approximately,” “about,” “substantially,” and the like mean within 10% (i.e., plus or minus 10%) of the recited value. Thus, for example, a recited angle of “about 80 degrees” refers to an angle ranging from 72 degrees to 88 degrees.

[0022] As described above, pressure vessel systems often include an emergency pressure relief valve (EPRV) coupled therewith for managing the internal fluid pressure of the pressure vessel to ensure the internal fluid pressure does not exceed a maximum operating pressure of the pressure vessel. In particular, the EPRV includes a cover or lid configured to move between a closed position blocking and sealing an opening in the pressure vessel to prevent fluid communication between the internal cavity of the pressure vessel and the outside environment, and an open position spaced apart from the opening in the pressure vessel to allow fluid communication between the internal cavity of the pressure vessel and the outside environment. Often, the cover is biased to the closed position with relatively heavy weights (e.g., 100 lbs.). In some cases, the cover of the EPRV and associated opening in the pressure vessel are relatively large, and thus, may be used by field personnel to access the internal cavity of the storage vessel for inspection, maintenance, and/or repairs. In cases where the cover of the EPRV is biased to the closed position by heavy weights, the weights and cover must be raised and lowered multiple times or the weights and cover must be disassembled and removed to access the opening and inside of the pressure vessel. The former approach (raising and lowering the cover and weights) is physically demanding and may present safety risks in the event the cover and weights are inadvertently dropped or slam shut, and the latter approach (disassembly and removing the weights and cover) are physically demanding and time consuming. Accordingly, embodiments described herein are directed to devices and methods for safely maintaining the cover of a pressure relief valve (e.g., EPRV) in an open position during inspection, maintenance, and/or repair operations in connection with the associated pressure vessel.

[0023] Referring now to FIGS. **1** and **2**, an embodiment of a fluid storage vessel **10** in accordance with principles disclosed herein is shown. In this embodiment, fluid storage vessel **10** generally includes a storage tank **20** and a pressure relief valve (e.g., EPRV) **50** coupled to the storage tank **20**. Storage tank **20** has a lower or bottom end situated on or proximal the ground (not shown) and an upper or top end **20***a* shown in FIG. **1**. In addition, storage tank **20** has an inner cavity (not shown) in which one or more fluids (e.g., hydrocarbon fluids) are contained and stored and an opening **21** in top end **20***a*. A fluid conduit **40** that extends from top end **20***a* and is in fluid communication with opening **21** and the inner cavity of storage tank **20**. The upper end of conduit

40 includes an annular tank flange **41**. A plurality of circumferentially-spaced holes extend through tank flange **41**. Conduit **40** and opening **21** provide access to the internal cavity of storage tank **20**. Although storage tank **20** is generally described as being vertically oriented, it may be understood that the configuration of storage tank **20** may vary in other embodiments from that shown in FIGS. **1** and **2**.

[0024] Pressure relief valve **50** is coupled to storage vessel **10** via tank flange **41** and provides controlled, selective access to and fluid communication between the internal cavity of storage tank **20** and the external environment. Pressure relief valve **50** includes an annular attachment flange or plate **51** fixably secured to mating tank flange **41** and a valve member **80** pivotably coupled to attachment plate **51**. Attachment plate **51** includes a central opening **52** and a plurality of uniformly circumferentially-spaced holes **53** disposed about central opening **52**. Each hole **53** is aligned with a corresponding hole in tank flange **41**. A bolt **54** extends through the aligned holes in the tank flange **41** and attachment plate **51**, and nuts are threaded onto bolts **54** (above and below flange **41** and plate **51**) to secure attachment plate **51** to tank flange **41**, and sealingly compress attachment plate **51** against tank flange **41**, thereby coupling pressure relief valve **50** to the tank flange **41**.

[0025] In this embodiment, a pair of laterally spaced, parallel anchoring projections or braces **56** extend vertically upward from attachment plate **51**. In particular, each brace **56** has a lower end **56***a* fixably secured to plate **51** and an upper end **56***b* opposite lower end **56***a* and distal plate **51**. Each anchoring brace **56** includes a throughbore **57** proximal upper end **56***b* for slibably and rotatably receiving a pivot pin **58**.

[0026] Referring still to FIGS. **1** and **2**, valve member **80** includes an elongate pivot arm **81** pivotally coupled to attachment plate **51**, a cover or lid **82** fixably attached to arm **81**, and a heavy weight **84** coupled to arm **81**. Lid **82** is configured to be selectively moved between an open position spaced apart from attachment plate **51** and a closed position sealingly engaging attachment plate **51**. In the closed position, lid **82** closes, blocks, and seals central opening **52** in attachment plate **51**, thereby preventing fluid communication between the internal cavity of storage tank **20** and the outside environment via opening **52** and fluid conduit **40**. However, in the open position, lid **82** is spaced apart from attachment plate **51**, thereby allowing fluid communication between the internal cavity of storage tank **20** and the outside environment via opening **52** and fluid conduit **40**.

[0027] Heavy weight **84** biases cover **82** to the closed position. In particular, pivot arm **81** has a first end **81***a* and a second end **81***b* opposite end **81***a*. A throughbore extends horizontally through pivot arm **81** proximal end **81***a*. Pivot arm **81** is positioned between braces **56** with the throughbore in pivot arm **81** coaxially aligned with throughbores **57** in braces **56**, and pivot pin **58** extends through the throughbore in pivot arm **81** and throughbores **57** in braces **56**, thereby pivotally coupling pivot arm **81** to attachment plate **51**. In this embodiment, the throughbore in pivot arm **81** and throughbores **57** in braces **56** are positioned and oriented such that pivot pin **58** extends horizontally therethrough (i.e., pivot pin **58** is horizontally oriented). Cover **82** is attached to pivot arm **81** proximal end **81***b* and heavy weight **84** is attached to pivot arm **81** above cover **82** proximal end **81***b*. As a result, heavy weight **84** biases cover **82** downward to the closed position (via gravity) as pivot arm **81** is free to pivot relative to attachment plate **51** about pin **58**.

[0028] As previously described, cover **82** has an open position and a closed position, and is biased to the closed position by heavy weight **84**. Cover **82** is part of valve member **80** of pressure relief valve **50**, and thus, valve member **80** and pressure relief valve **50** may also each be described as having an open position with cover **82** spaced apart from attachment plate **51** and a closed position with cover **82** sealingly engaging attachment plate **51**.

[0029] As described above, cover **82** (and hence valve member **80** and pressure relief valve **50**) are normally biased to the closed position by heavy weight **84**. In the closed position, relief valve **50** restricts and/or prevents fluid communication between the internal cavity of storage tank **20** and the external environment surrounding storage tank **20**, such that the internal cavity is sealed from the external environment. Conversely, in the open position, relief valve **50** permits fluid communication between the internal cavity of storage tank **20** and the external environment surrounding the storage tank **20** such that fluid may flow between the internal cavity of storage tank **20** and the external environment. However, valve member **80**, and hence valve member **80** and pressure relief valve **50**, transition or actuate from the closed position to the open position in response to fluid pressure within the internal cavity of storage tank **20** reaching a predefined threshold pressure that exceeds the biasing force provided the heavy weight **84**, such as a maximum operating pressure of the storage tank **20** that is based on the configuration and design of the storage tank **20**. For example, fluid vapor may collect within the internal cavity of storage tank **20** and increase in pressure within storage tank **20** until the fluid pressure within the internal cavity within storage tank **20** reaches the threshold pressure. Upon reaching the threshold pressure, the fluid pressure in the internal cavity of storage tank **20** exceeds the biasing force exerted by the heavy weight **84**, thereby urging cover **82** upward and lifting cover **82** from attachment plate **51** to transition cover **82** to the open position and allow relief of the excess fluid pressure within the internal cavity of storage tank **20**. Once sufficient fluid pressure within the internal cavity of storage tank **20** is relieved, the biasing force applied by heavy weight **84** exceeds the fluid pressure in the internal cavity of storage tank **20**, thereby urging cover **82** downward to the closed position.

[0030] As previously described, heavy weight **84** biases cover **82** and valve member **80** to the closed position. To access the internal cavity of storage tank **20** for inspection, maintenance, repair, or combinations thereof, cover **82** and valve member **80** must be moved to the open position. This can be achieved by removing heavy weight **84**, or manually lifting cover **82** and heavy weight **84**, and then holding cover **82** and heavy weight **84** in a position sufficiently spaced above attachment plate **51** to allow access to the internal cavity of storage tank **20**. Removing heavy weight **84** from pivot arm **81** (and then reattaching heavy weight to pivot arm **81** after the inspection, maintenance, and/or repair) is time consuming; and manually lifting and holding cover **82** and heavy weight **84** is hazardous and potentially dangerous. Accordingly, embodiments described herein are directed to devices, which may be referred to herein as “hand saver” devices, for safely maintaining a pressure relief valve (e.g., pressure relief valve **50**) in an open position during the inspection, maintenance, and/or repair of storage tank **20**.

[0031] Referring now to FIGS. **3** and **4**, an embodiment of a hand saver device **100** for safely maintaining cover **82**, and hence valve member **80** and pressure relief valve **50**, in an open position is shown. In this embodiment, hand saver device **100** includes a body **110** and an elongate locking rod **130** removably coupled to body **110**. In this embodiment, one or more fasteners, for example, a locking pin **140** is removably coupled to locking rod **130** to prevent locking rod **130** from inadvertently decoupling from body **110**.

[0032] Referring now to FIGS. **3-7**, in this embodiment, body **110** includes a generally U-shaped base plate **112** and a pair of parallel, laterally spaced extension arms or plates **118** extending upwardly and perpendicularly from base plate **112**. More specifically, as best shown in FIGS. **5-7**, base plate **112** is a generally rigid plate having a central axis **115**, a first end **112***a*, a second end **112***b* opposite end **112***a*, a pair of lateral sides **113** extending axially between ends **112***a*, **112***b*, and a recess **114** extending axially (relative to central axis **115**) from first end **112***a* toward but not to second end **112***b*. In some embodiments, base plate **112** includes a pair of openings **111** proximal end **112***b* and lateral sides **113** for connecting a pair of chains or pull cords **142** to base plate **112**. As best shown in FIGS. **3** and **4**, each pull cord **142** has one end coupled to base plate **112** via a corresponding opening **111** and an opposite end removably coupled to an end of locking rod **130** via a split ring **146** (removably disposed through one end of locking rod **130**) or locking pin **140** (removably attached to the opposite end of locking rod **130**).

[0033] Referring again to FIGS. **5-7**, recess **114** is laterally centered (relative to central axis **115**) along end **112***a*, and may be described as defining a pair of elongate, parallel, plate-shaped members **116** extending axially (relative to central axis **115**) from end **112***a* on opposite lateral sides of recess **114**. Each elongate plate-shaped member **116** includes a throughbore or hole **120** proximal end **112***a* for securing hand saver device **100** to attachment plate **51** of pressure relief valve **50**. In some embodiments, elongate plate-shaped members **116** may also include a pair of parallel reinforcement, support plates **117** coupled thereto on opposite lateral sides of recess **114**.

[0034] Extension arms **118** extend perpendicularly from the upper surface of base plate **112**, and in particular, from elongate plate-shaped members **116**. Arms **118** are generally oriented vertically and parallel to central axis **115**, and are disposed on opposite sides of recess **114**. Each extension arm **118** may be described as having a first or fixed end **118***a* fixably secured to base plate **112** and a second or free end **118***b* distal base plate **112**. Each extension arm **118** includes a throughbore or hole **119** extending therethrough between ends **118***a*, **118***b*. Holes **119** are coaxially aligned and sized to slidably receive locking rod **130** as shown in FIGS. **3** and **4**.

[0035] As described above, the hand saver device **100** is used to safely maintain the pressure relief valve **50** in an open position during inspections, maintenance, repair, or combinations thereof. In particular and as best shown in FIG. **4**, hand saver device **100** is removably coupled to attachment plate **51** by horizontally orienting base plate **112** with extension arms **118** extending vertically upward, positioning pivot arm **81** and braces **56** in recess **114** between with elongate plate-shaped members **116** (i.e., with plate-shaped members **116** disposed on opposite lateral sides of pivot arm **81** and braces **56**), and holes **120** coaxially aligned with corresponding holes **53** in attachment plate **51**. Next, base plate **112** is bolted to attachment plate **51** via bolts **54** that are passed through aligned holes **53**, **120** to fixably secure base plate **112** to attachment plate **51**. Next, cover **82** is lifted to transition cover **82**, valve member **80**, and relief valve **50** to the open positions. As cover **82** is raised, end **81***a* of pivot arm **80** rotates downward within recess **114** generally below holes **119** of extension arms **118**. With end **81***a* positioned vertically below holes **119**, locking rod **130** is advanced through aligned holes **119**, and then secured within holes **119** via locking pin **140** and split ring **146**, which are disposed through holes in opposite ends of locking rod **130**, to prevent locking rod **130** from being inadvertently pulled or removed from holes **119**. Next, cover **82** is carefully lowered until end **81***a* of pivot arm **80** directly engages and bears against locking rod **130**, thereby preventing cover **82** from dropping further and transitioning to the closed position.

[0036] In this manner, the hand saver device **100** is connected to the pressure relief valve **50** with cover **82** and relief valve **50** in the open position to safely allow access to the internal cavity of tank vessel **20** for inspection, maintenance, repair, or combinations thereof. As described, heavy weight **84** does not need to be removed, and further, relief valve **50** does not need to be removed, thereby facilitating a safe and more efficient manner of maintaining cover **82** and relief valve **50** in an open position. After the inspection, maintenance, and/or repair, hand saver device **100** may be disengaged from the pressure relief valve **50** by generally performing the installation process in reverse.

[0037] While preferred embodiments have been shown and described, modifications thereof can be made by one skilled in the art without departing from the scope or teachings herein. The embodiments described herein are exemplary only and are not limiting. Many variations and modifications of the systems, apparatus, and processes described herein are possible and are within the scope of the disclosure. For example, the relative dimensions of various parts, the materials from which the various parts are made, and other parameters can be varied. Accordingly, the scope of protection is not limited to the embodiments described herein, but is only limited by the claims that follow, the scope of which shall include all equivalents of the subject matter of the claims. Unless expressly stated otherwise, the steps in a method claim may be performed in any order. The recitation of identifiers such as (a), (b), (c) or (1), (2), (3) before steps in a method claim are not intended to and do not specify a particular order to the steps, but rather are used to simplify subsequent reference to such steps.

Claims

1. A hand saver device for maintaining a relief valve in an open position, the relief valve having a closed position with a lid of the relief valve closing an opening of a storage tank and the open position with the lid pivoted away from the opening of the storage tank, the hand saver device comprising: a body comprising a U-shaped base plate having a central axis, a first end, a second end opposite the first end, a pair of lateral sides extending axially from the first end to the second end, and a recess extending axially from the first end; a pair of laterally spaced extension arms extending upwardly from the base plate, wherein each extension arm includes a throughbore extending laterally therethrough; and an elongate locking rod configured to be removably and slidably disposed in the throughbores of the extension arms.
2. The device of claim 1, further comprising a locking pin configured to be removably disposed in a hole extending through the locking rod to prevent the locking rod from decoupling from the extension arms.
3. The device of claim 1, wherein the recess is configured to at least partially receive a pivot arm of the relief valve coupled to the lid when the relief valve is in the open position.
4. The device of claim 3, wherein the base plate includes a pair of elongate, parallel, plate-shaped members disposed on opposite sides of the recess.
5. The device of claim 4, wherein the base plate includes a first throughbore extending through one of the plate-shaped members proximal the first end of the base plate and a second throughbore extending through the other of the plate-shaped members proximal the first end of the base plate, wherein each throughbore of the base plate is configured to a receive a fastener to fixably couple the base plate to the storage tank.
6. The device of claim 3, wherein the locking rod is configured to engage the pivot arm of the relief valve.
7. A fluid storage vessel, comprising: a storage tank having an internal cavity configured to store one or more fluids; a pressure relief valve coupled to the storage tank and having an open position and a closed position, wherein the pressure relief valve comprises; an annular attachment plate fixably secured to the storage tank, wherein the attachment plate includes an opening in fluid communication with the internal cavity of the storage tank; a valve member including a pivot arm pivotally coupled to the attachment plate and a cover fixably attached to the pivot arm, wherein the cover engages the attachment plate and closes the opening of the attachment plate with the pressure relief valve in the closed position, and wherein the cover is spaced apart from the attachment plate and opens the opening of the attachment plate with the pressure relief valve in the open position; and the hand saver device of claim 1, wherein the base plate of the body is removably coupled to the attachment plate of the pressure relief valve and the locking rod engages the pivot arm to maintain the pressure relief valve in the open position.
8. The storage vessel of claim 7, wherein the base plate is bolted to the attachment plate.
9. A method for maintaining a relief valve coupled to a storage tank in an open position, wherein the relief valve including a pivot arm having a first end and a second end, a weight attached to the pivot arm proximal the first end, and a lid coupled to the pivot arm proximal the first end, the method comprising: (a) coupling a hand saver device to the relief valve with the relief valve in a closed position, wherein the hand saver device comprises: a body comprising a base plate and a pair of extension arms extending upwardly from the base plate; wherein the base plate has a central axis, a first end, a second end opposite the first end, a pair of lateral sides extending axially from the first end to the second end, and a recess extending axially from the first end; wherein the extension arms are disposed on opposite lateral sides of the recess, and wherein each extension arm includes a throughbore; an elongate locking rod configured to be removably disposed in the throughbores of the extension arms; (b) raising the lid to transition the relief valve from the closed position to the open position and move the second end of the pivot arm downward through the recess of the base plate after (a); and (c) advancing a locking rod through the throughbores of the

extension arms after (b) to prevent the second end of the pivot arm from rotating upward through the recess.
10. The method of claim 9, wherein (a) comprises fixably coupling the base plate of the hand save device to an annular attachment plate of the pressure relief valve.
11. The method of claim 9, further comprising (d) removably coupling a fastener to an end of the locking rod after (c) to prevent the locking rod from being removed from the throughbores of the extension arms.
12. The method of claim 11, wherein the fastener comprises a locking pin.
13. The method of claim 9, further comprising (d) lowering the lid after (c) until the second end of the pivot arm directly engages the locking rod.
14. The method of claim 9, wherein the base plate comprises a pair of elongate, parallel plate-shaped members disposed on opposite lateral sides of the recess, and wherein the plate-shaped members are disposed on opposite sides of the pivot arm after (a).
15. The method of claim 14, wherein each of the plate-shaped members comprises a throughbore, and wherein (a) comprises bolting the base plate to the of the hand save device to an annular attachment plate of the pressure relief valve using the throughbores of the plate-shaped members.
